Samrat Tidke
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Professional Summary

A Senior Data Engineer and MLOps expert with over 15 years of experience, specializing in designing scalable data pipelines and implementing end-to-end MLOps solutions across multiple cloud platforms (AWS, GCP, Azure). Proficient in DevOps practices, containerization, and managing AI/ML lifecycle deployments using cutting-edge tools such as Kubernetes, Docker, and CI/CD pipelines. Skilled at integrating AI/ML models into production environments and optimizing big data workflows, with a strong combination of data engineering (60%) and MLOps (40%) expertise. Demonstrates a successful career transition from Mechanical Engineering to Data Analytics, leveraging engineering insights to drive data-driven solutions and optimize processes.

Technical Skills

- **Data Engineering (40%):** Spark 3.0, PySpark SQL, BigQuery, Airflow, Kafka, Superset, MinIO
- **MLOps (30%):** MLflow, Docker, FastAPI, TensorFlow, Keras, Kubernetes, Jenkins, GitLab, CI/CD, Seldon Core, Evidently AI, GPU
- **Scrum (20%):** Scrum, Jira, Stories for MLOps Architecture
- **Programming:** Python, SQL, Java
- **DevOps & Cloud:** Kubernetes, Docker, Jenkins, AWS, GCP, Azure
- **Machine Learning:** Topic Modelling, Regression, Classification, SVM, Clustering, ANN, RNN, LSTM, GPU

Experience

Eviden (Atos India)
Senior Consultant | Dec 2021 – Present

- **Credit One Bank:**
  - Led the design and implementation of data pipelines supporting Okta, CIAM, and customer account opening processes using Kafka, PySpark SQL, and MinIO.
  - Collaborated with Analytics teams to integrate pipelines using Kubernetes, Docker, and GitLab CI/CD for automated deployment and monitoring.
  - Ensured data security and compliance when incorporating CIAM into workflows, optimizing data handling, and improving privacy adherence.
- **Renom:**
  - Managed ETL pipelines for IoT data, focusing on real-time data ingestion and processing using AWS services (S3, kafka, EC2).
  - Implemented LSTM models for predictive maintenance, significantly reducing downtime for industrial equipment.
  - Directed cloud-agnostic deployments, leveraging AWS infrastructure to optimize resource use and improve system scalability.
  - Monitored scripts on Evidently AI and Airflow triggers for retraining.
- **NAM:**
  - Developed and maintained NLP pipelines using Azure, focusing on text data extraction, cleaning, and transformation.
  - Automated the deployment of machine learning models using CI/CD pipelines, enhancing the reliability of model updates on Azure Notebooks.
  - Applied topic modelling techniques to analyze large unstructured datasets, driving key business insights.

- **J&J:**
    - Designed end-to-end data pipelines from ingestion to feature engineering for machine learning model development.
    - Deployed machine learning models using Kubernetes, Kubeflow, and Seldon Core, ensuring scalability and performance.
    - Implemented GPU programming for AI model acceleration and streamlined deployment through Docker, Kubernetes, and Jenkins-based CI/CD pipelines.
    - Deployed models using Seldon Core servers.

Productiva IT Solutions

GCP Data Engineer | April 2021 – Oct 2021

- **Palo Alto Networks:**
    - Analyzed and optimized ETL pipelines within Google Cloud Platform (GCP), automating workflows with Airflow and Jenkins to enhance reliability and performance for cybersecurity products.
    - Developed custom monitoring scripts using PySpark and Python to ensure data quality and compliance with governance standards critical to cybersecurity operations.
    - Provided ongoing support for data engineering activities, maintaining high availability and performance of ETL pipelines deployed using Kubernetes and Docker.
    - Implemented version control and collaboration processes with GitHub, facilitating efficient CI/CD practices and seamless integration across teams.
    - Managed secure data transfers via FTP protocols, ensuring the integrity and security of data supporting laptop and computer cybersecurity solutions.

IAV India Pvt Ltd

Data Analyst | Jul 2017 – Dec 2020

- **Predictive Analytics:**
    - Conducted Proof of Concepts (POCs) for data ingestion using Kafka connectors and PySpark, optimizing data loading into Hive and Cassandra tables.
    - Developed and maintained robust data pipelines for real-time sensor data, facilitating predictive maintenance and reducing downtime.
- **Data Analytics:**
    - Analyzed data from engine test benches using proprietary IAV DE software to identify trends and insights.
    - Utilized Python and R libraries for comprehensive data analysis, visualization, and the development of predictive models.
    - Integrated advanced predictive models to enhance early fault detection accuracy, optimizing maintenance workflows and reducing costs.
- **Exploratory Data Analysis (EDA):**
    - Conducted EDA to uncover patterns and correlations in large datasets, informing predictive model development and business strategies.
- **Vehicle Projects for VW Group:**
    - Led data analytics initiatives supporting vehicle component design, optimizing project timelines and resource allocation.
    - Managed teams focused on the development and analysis of different vehicle parts, ensuring data-driven decision-making throughout the project lifecycle.
    - Collaborated with engineering teams to align data analytics efforts with design and development objectives, fostering a cohesive project environment.
- **Transition to Data Analytics:**
    - Identified the potential of data-driven approaches to improve vehicle part design and maintenance processes.
    - Completed internal training programs in data analytics, ETL processes, and relevant programming languages (Python, R).

    - Initiated collaboration with the data analytics team to integrate engineering insights with data-driven strategies.

Mechanical Engineer | Feb 2009 – Jun 2017

- **Vehicle Part Design:**
    - Designed and developed vehicle components using advanced CAD software, ensuring compliance with VW Group standards and specifications.
    - Collaborated with cross-functional teams, including design, testing, and manufacturing, to streamline the vehicle part development process.
    - Managed project timelines, resources, and budgets for multiple vehicle component projects, enhancing efficiency and delivery.
    - Conducted feasibility studies and implemented design improvements based on testing feedback and performance data.
    - Utilized proprietary IAV DE software to simulate and validate vehicle part performance under various conditions.
- **Transition to Data Analytics:**
    - Identified the potential of data-driven approaches to improve vehicle part design and maintenance processes.
    - Completed internal training programs in data analytics, ETL processes, and relevant programming languages (Python, R).
    - Initiated collaboration with the data analytics team to integrate engineering insights with data-driven strategies.

Magna Steyr India Pvt Ltd
Engineer | Jan 2008 – Feb 2009

- Coordinated offshore engineering activities, focusing on data management and reporting to ensure project alignment with client expectations.
- Developed and managed databases to track project progress, resource allocation, and deliverables.
- Analyzed project data to identify potential bottlenecks and implemented solutions to keep the project on track.

Mungi Engineers Pvt Ltd
Development Engineer | June 2007 – Jan 2008

- Optimized manufacturing processes by analyzing production data and identifying areas for improvement.
- Implemented quality control measures based on data analysis, leading to a reduction in defects and improved product quality.
- Collaborated with production teams to refine process workflows, increasing efficiency, and reducing waste.

Seam Arc Welding Products Pvt. Ltd.
Executive Engineer | May 2005 – June 2007

- Developed data-driven solutions to optimize manufacturing processes, focusing on improving productivity and reducing operational costs.
- Implemented data collection and analysis systems to monitor production performance and identify opportunities for improvement.

Key Achievements

- **Career Transition:** Successfully transitioned from a Mechanical Engineering role to a Data Analyst position within **IAV India Pvt Ltd**, leveraging engineering expertise to enhance data-driven projects.

- **Process Optimization:** Implemented data pipelines that reduced data processing time by 30%, enabling faster decision-making for vehicle maintenance.
- **Predictive Modeling:** Developed predictive models that increased early fault detection accuracy.

Skills

- **Engineering:** CAD Design, Vehicle Component Development, Project Management, Mechanical Testing
- **Data Analytics:** Python, R, PySpark, Kafka, Hive, Cassandra, Data Visualization, Predictive Modeling, ETL Processes
- **MLOps & Data Engineering:** Spark, BigQuery, Airflow, Superset, MinIO, MLflow, FastAPI, TensorFlow, Keras, Kubernetes, Jenkins, GitLab, Seldon Core, Evidently AI, GPU Programming
- **DevOps & Cloud:** Kubernetes, Docker, Jenkins, AWS, GCP, Azure
- **Tools:** IAV DE Software, Tableau, Power BI, SQL, Jira
- **Scrum:** Scrum Framework, Agile Methodologies

Education

- **Bachelor of Engineering in Mechanical Engineering**
  K.K. Wagh College of Engineering, Nashik | 53%
- **Higher Secondary Certificate (HSC)**
  K.T.H.M. College, Nashik | 63.5%
- **Secondary School Certificate (SSC)**
  RJEMS High School, Nashik | 65.86%

Place: Pune | **Date:** 2024-04-14
Signature: Samrat Tidke